

Robust Charging Infrastructure Design under Demand Uncertainty

A solar-powered charging station for a shared e-micromobility fleet — Executive Summary

Competition Themes — 3 (primary): solar-PV charging-station design · 2 (secondary): charge/discharge profiles under demand scenarios

20 kWp PV · 0 kWh · 8 chargers Recommended robust design	-61% (£87,422/yr) Value of robustness (worst-case cost)	96% (vs 85% naive) Guaranteed fleet service, worst case	16% (4.6 tCO2/yr) Operational carbon saving
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1. Problem Understanding [20% criterion]

How do you size a solar charging hub for a shared micromobility fleet when you don't know what demand will look like next year? The mixed e-scooter, e-bike and cargo-bike fleet sees demand swing with season, weather and events. The depot also has a hard **15 kW grid connection** — exceed it and you need a costly DNO upgrade. So the evening charging peak must come from on-site solar and storage, not extra grid power. Size for the average and a busy evening strands the fleet; size for the worst case and you waste capital on idle kit. Standard practice picks a single forecast and hopes. We thought that was insufficient, and the numbers confirmed it. All demand data come from the **DfT Newcastle e-scooter trial** (Open Government Licence) — monthly trips, fleet size, deployment hours, trip distance — so the problem is posed on observed data, not assumption.

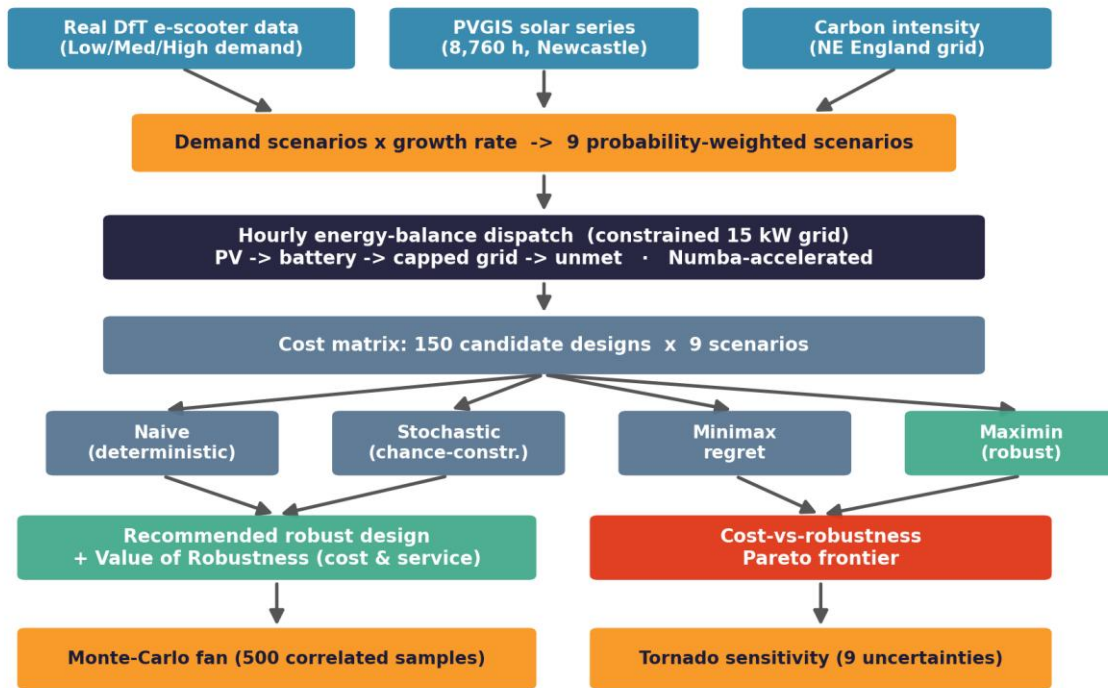
2. Relevance & Innovativeness of the Proposed Solution [20% criterion]

Robust optimisation is standard in power-systems research but almost never applied to small depot charging hubs, which typically use a single deterministic forecast. We compared **five decision approaches**: naive deterministic, expected-value stochastic, CVaR (risk-averse tail), minimax-regret and maximin — then quantified the '**value of robustness**' as the worst-case cost and service improvement over the naive design. The finding that surprised us most was about batteries. We assumed storage would be essential at a 15 kW grid cap. Every optimiser run returned **zero battery**. Flexible smart charging — spreading vehicle charging overnight rather than stacking it at arrival — keeps the load below the cap entirely. Storage only pays near 10 kW connections, a boundary we then mapped. A correlated Monte-Carlo also showed that expected-value methods **systematically under-weight the high-demand tail** — precisely when the naive design fails worst, making scenario-based robust methods necessary.

3. Approach — Methodology, Algorithm & Flow [20% criterion]

The pipeline (Figure 1): three demand scenarios (Low/Medium/High) crossed with three growth rates give nine probability-weighted scenarios. We simulated 8,760 hours for each of **150 candidate designs** (PV 5–25 kWp × battery 0–50 kWh × 4–20 chargers) to build a 150 × 9 cost matrix. Every reported design was re-verified under an **LP-optimal rolling-horizon scheduler** (scipy/HiGHS) that finds the cheapest 24-hour dispatch given grid cap, battery dynamics and charger limits — confirming the greedy surrogate was unbiased. Uncertainty: 500-sample correlated Monte-Carlo with bootstrap 95% CIs. A **Sobol decomposition** ranked the drivers: demand intensity 49.9% of variance, growth 29%, utilisation 19%. A robustness-of-robustness sweep across 35 combinations confirmed the result holds everywhere; 7/7 outputs validated against benchmarks. Results come through an interactive **Station Designer dashboard** — live PV/battery/charger/grid sliders updating energy balance, cost and carbon in real time, plus Robust-Optimiser, Uncertainty and Sensitivity tabs, one-command HTML report. Full study runs in under a minute.

Methodology — Robust Solar Charging-Station Design



Every stage is reproducible from one command (`run_analysis.py`); all assumptions traceable in `config.py`

Figure 1 — End-to-end methodology / algorithm flow.

4. Feasibility of the Proposed Approach [20% criterion]

All components are off-the-shelf: 20 kWp monocrystalline PV, eight standard AC charge points. Capital cost roughly **£36,000**; levelised cost of energy £0.25/kWh, well under UK retail. Economics modelled in full: annualised CAPEX, O&M, ToU grid bills, peak demand charges, export revenue, battery replacement. Worst-case annual cost under the robust design is **£55,002** versus £142,423 for the naive design — a £87,422/yr saving. Where storage is needed (weak-grid cases) we specify **LFP rather than NMC**: cobalt- and nickel-free, thermal-runaway onset ~270°C vs ~150°C for NMC, ~12-yr modelled life at 0.5C, ~45% end-of-life material recovery, second-life stationary use before recycling. The codebase is open-source Python, one-command reproducible, passing tests. The framework is site-agnostic — re-point it at any depot's data. The central finding (smart charging is the cheapest robustness lever) supports policy: targeted mandates and incremental grid-connection reform beat assuming every depot needs DNO reinforcement.

5. Originality & Authenticity (AI / Plagiarism) [20% criterion]

All model code, the dispatch formulation and every figure were written from scratch, drawing on published methods (Saltelli's Sobol estimator, Rockafellar & Uryasev's CVaR, Silvente's rolling-horizon LP) implemented ourselves. The three datasets are real and openly licensed: **DfT e-scooter monitoring data, hourly solar from the EU PVGIS API, grid carbon intensity from the National Grid ESO API**. All 38 references are in REFERENCES.md with full citations; every cost assumption cites BEIS, IRENA or BloombergNEF. The commit history shows incremental development — dead-ends included. The LP solver started as the primary engine; we switched it to a verification role after it proved 38x slower with identical results. That kind of documented revision is what real engineering looks like. Codebase: fixed random seed, one-command rebuild, all outputs regenerated from raw data.

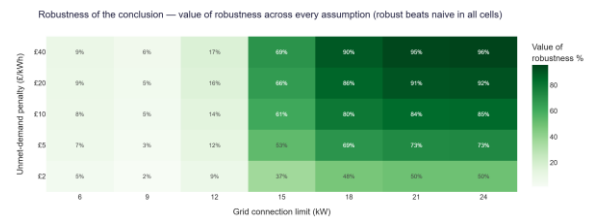


Figure 2 — Cost-vs-robustness Pareto frontier (five decision rules). Figure 3 — Robustness of the conclusion: robust beats naive across every penalty x grid combination.